Review 2 prep guide: implementation in depth

Forecasting hidden visual-grounding regression in VLM fine-tuning. Feasibility pilot (Tiers 1-3). All numbers come from the saved results. Feasibility sketch, not evidence.

0. The story in 30 seconds

When you fine-tune a vision-language model, it can quietly start relying less on the image while its overall score stays the same. Our main goal is to **forecast that early in training**. Before building a forecaster, we ran a feasibility pilot to check that the target can be measured. It can: on InternVL3, overall accuracy stayed flat while image benefit on image-necessary questions dropped about 4 points, consistently across seeds. The forecaster is the next phase.

1. The problem and why it matters

- **Vision-language models** (VLMs) are often fine-tuned on new data.
- Teams check that **overall accuracy** didn't drop and assume all is well.
- But a model can get answers right **without looking at the image**, using language patterns or common sense. If fine-tuning pushes it further that way, accuracy can stay flat while **visual grounding** gets worse. We call this a **hidden regression**.
- **Why it matters:** in real use (medical images, documents, robots), a model that stops looking breaks on unusual cases that accuracy benchmarks don't catch.
- **Supporting papers:** *Seeing without Looking* (accuracy barely moves when image tokens are removed), *Words or Vision* (VLMs trust text over images), and Dancette et al. (shortcuts hide behind accuracy).

2. Why a pilot first, not the forecaster

- A forecaster predicts a **target**. If the target can't be measured reliably, the forecaster has nothing to learn from.
- The pilot asked three questions. **Tier 1:** can we measure image reliance cleanly? **Tier 2:** can we build training data that should push the model toward or away from the image, without leakage? **Tier 3:** does fine-tuning on that data move image reliance by a measurable amount?
- **Answer:** yes, yes, and yes at about 2-4 points, depending on the model.

3. Every design decision: what, why, and the alternatives

3.1 Models

Model	Size	Precision	Why
Qwen2-VL-2B	~2B	4-bit (QLoRA)	Strong small open model, widely used; fits on the 8 GB laptop GPU in 4-bit.
SmolVLM-2.2B	~2.2B	bf16	A different design (SigLIP vision encoder, SmolLM2 language model). 4-bit dropped its MMStar accuracy from 40% to 29%, so we kept bf16.
InternVL3-2B	~2B	bf16	A third family and the strongest base model (56.8% on MMStar). bf16 keeps it comparable with SmolVLM.

- **Why three models:** one model can't show whether the effect is general or a quirk. Three families have different vision encoders and connectors.
- **Why ~2B and not 7B:** the hardware is an **RTX 4060 Laptop GPU with 8 GB of memory**. 2B models with LoRA fit (peak 2.1-5.5 GB). 7B would need more memory or cloud compute.
- **Why open models:** you can't LoRA fine-tune GPT-4o or Claude, and you can't read their internals (loss, gradients, attention).
- **Image budget kept similar:** Qwen and InternVL3 both use about 256 visual tokens per image (InternVL3 is set to one 448 px tile). SmolVLM uses up to about 405.

3.2 Evaluation benchmark: MMStar

- **What it is:** 1,500 multiple-choice questions built specifically so the image is needed. The authors filtered out questions answerable from text alone.
- **Why it suits us:** if questions could be answered without the image, image reliance couldn't be measured on them.
- **Check:** Qwen got 26.4% with a gray image, close to the 25% chance level. So the benchmark really does need the image.
- **Why 500 items for the A set:** a balance between measurement precision and evaluation time on the laptop.

3.3 Scoring: letter log-probability

- We give the model the image, question and options A-D, then read the probability it gives to each letter as the **next token**. The highest one is the answer.
- **Why not free text generation:** no answer parsing (so no "the model said 'Option B is...'" mistakes); deterministic (the same input always gives the same answer); one forward pass per question, so it's fast; and it gives a confidence for every option.
- **SmolVLM detail:** it wants to start its reply with "Answer", so we add " Answer:" before reading the letter. Without that, 98% of its probability went to the word "Answer".
- **Option shuffling (Tier 1):** we also ran 5 shuffled option orders to check for **letter bias**, a preference for A regardless of content.

3.4 Three image conditions

Condition	What	Why
Real	The correct image	Normal performance
Blank	Gray image (value 127), same size	Removes all visual information but keeps the format and number of image tokens, so the model still "sees" an image slot
Swap	Another question's image	Tests whether the model reacts to *wrong* visual content, not just missing content. A gray image is unnatural; a real but wrong image is a harder test.

- **Image benefit G = accuracy(real) - accuracy(blank).** A high G means the model uses the image a lot.
- We report the swap version, accuracy(real) - accuracy(swap), separately as a second check.
- The swaps are fixed with a seed, so every model and checkpoint sees the same swapped image.

3.5 Two evaluation sets: A and G

- **A (500 items):** a random sample of MMStar. It measures **overall accuracy**, the number teams usually check.
- **G (581 items):** taken from the **other 999** MMStar items. We kept only questions Qwen got right **without** the image in at most 1 of 6 option orders, so they're questions you can't guess.
- **Why separate pools:** choosing G from the same items as A would mix up the two measurements.
- **Why both:** the hidden regression is **A flat while G drops**. That needs both numbers.
- **Known caveat:** G was chosen using **Qwen's** failures, so Qwen's blank score on G starts artificially low (3.8% vs 7.3% on fresh option orders) and drifts up after any training (regression to the mean). That's why we compare **text90 vs text10**, never one run against the base model. Both runs share the same bias, so it cancels.

3.6 Training data: The Cauldron

- A large public collection of training datasets for VLMs (from the Idefics2 work).
- **Subsets chosen:** VSR (spatial yes/no), A-OKVQA (knowledge), COCO-QA (objects), TallyQA (counting).
- **Why these four:** none of them is a source MMStar was built from (ScienceQA, AI2D, MathVista, MMBench, SEED, MMMU), so there's less risk of overlap. They also include both text-guessable and image-needing questions.
- We streamed 800 per subset and kept one question per image, giving 3,200. We dropped 33 duplicate images and 6 MMStar overlaps, leaving **3,161 items**.

3.7 Labels: the loss gap

- For each training item, the base model (Qwen) scores the correct answer twice: once with the real image and once with gray.
- **Loss gap = loss(gray) - loss(real).** Small gap: the answer is about as easy without the image, so the item is **text-answerable**. Large gap: the image really helps, so it's **image-necessary**.
- **Thresholds:** bottom 40% = text-answerable, top 60% = image-necessary. The middle 20% is dropped so the two classes are clearly separated.
- **Thresholds are set within each subset:** VSR naturally has different gaps from TallyQA. Global thresholds would make the label mostly mean "which dataset it came from" (a confound we found in the first version).
- **Why loss instead of right/wrong:** these are short free-text answers, and loss gives a continuous, more sensitive score.
- **Same idea as VisNec (2026),** which scores training samples the same way. VisNec uses it to select data; we use it to vary the mix.
- **Caveat:** labels came from Qwen and were reused for all three models, so they're Qwen's view of which items are text-answerable.

3.8 Two training mixes

- **text10:** 1,000 items, 10% text-answerable. Mean loss gap 1.17.
- **text90:** 1,000 items, 90% text-answerable. Mean loss gap 0.09.
- **Both have exactly 250 items from each subset.** This is the "matched" design, so the source mix can't explain any difference.
- **Hypothesis:** training mostly on items answerable without the image rewards the language shortcut, so image reliance drops.
- **Why 10% vs 90% and not 0% vs 100%:** strong contrast while keeping some of each type, which is more realistic.
- **Later finding:** the mixes still differed in **yes/no balance**. Among VSR answers, text10 had 75% "yes" and text90 had 36%. See section 4.4.

3.9 Leakage guard (hard rule)

- If test questions appear in training data, the results mean nothing.
- **Image check:** a 64-bit perceptual hash (dHash). Two images within 4 bits of each other count as the same, which catches resized or re-compressed copies.
- **Text check:** identical normalised question text of at least 30 characters. Shorter questions like "how many?" are too generic to count.
- It checks against **all 1,500 MMStar items**, not just the ones we use, and also MMVP and NaturalBench.
- If anything overlaps, the run **stops with an error**. tests/test_overlap.py plants a duplicate to prove the guard fires.
- **Result:** 0 overlaps in both mixes. Two NaturalBench samples were flagged and dropped even though they looked like false positives, to be safe.

3.10 Fine-tuning: LoRA

- **What:** the model is frozen and small trainable "adapter" matrices are added next to its layers.
- **Where:** all 7 linear layers in every block of the **language model**: attention (q, k, v, o) and MLP (gate, up, down). There are 196 adapters for Qwen and InternVL3 and 168 for SmolVLM.
- **Vision encoder and connector are frozen.** The image features stay identical before and after training, so any change in image use must come from the language side **using** them differently. That makes the result easy to interpret.
- **Trainable parameters:** about 18 million, under 1% of the model.

Setting	Value	Why
Rank	16	Common default; not tuned
Alpha	32	Scale = alpha/r = 2, the usual setting
Dropout	0.05	Light regularisation
Learning rate	2e-4	Standard for LoRA, with warm-up and cosine decay

Setting	Value	Why
Batch	8	Via gradient accumulation, because of memory
Gradient clipping	1.0	Stability
Steps	120 (Qwen) / 125 (others)	About one pass over the 1,000 items (1000 / 8 = 125)

- **Loss on answer tokens only,** so the model isn't trained to repeat the question.
- **Why LoRA and not full fine-tuning:** full fine-tuning doesn't fit in 8 GB. LoRA is also what most practitioners use, so it's the realistic setting. The *Fine-tuning MLLMs Without Forgetting* paper uses LoRA too.
- **Cost per run:** 12-21 minutes, with peak memory 2.1 GB (Qwen) to 5.5 GB (SmolVLM).

3.11 Seeds and runs

- **3 models x 2 mixes x 3 seeds = 18 runs.**
- Seeds change the data order and the LoRA initialisation. Three seeds show whether the direction is consistent.
- **Caveat:** Qwen's seed-0 run was trained before LoRA initialisation was seeded. Seeds 1 and 2, and all SmolVLM and InternVL3 runs, are fully seeded.

3.12 Signals logged during training (the future forecaster's inputs)

- **Training loss**, separately for text-answerable and image-necessary items.
- **Gradient norm** at every step.
- **Attention on image tokens** for held-out items. This is supporting evidence only, not used in any claim.
- **Saved checkpoints:** 5 evaluated for Qwen; 3 for SmolVLM and InternVL3 (step 0, step 50 and the final step).

3.13 Paired-image benchmarks

- **MMVP:** 150 pairs of images that look alike but differ in one detail, with the same question for both (300 questions).
- **NaturalBench:** 298 samples, 1,192 questions.
- **Metric:** how often the model gives the **same answer to both images** when the right answers differ. A high value suggests it's ignoring the image.
- **Why add them:** they're a second, independent way to measure image use, not tied to MMStar.
- **Flaw we found:** a model that always says "Yes" also gives the same answer to both images. This metric mixes up **ignoring the image** with **answer bias**.

3.14 Statistics

- **Main comparison (the contrast):** (text90 change from step 0) - (text10 change from step 0) at the final step.
- **Why compare the two runs:** both runs share generic fine-tuning effects such as format learning and drift. Subtracting one from the other isolates the effect of the **data mix**.
- **Bootstrap 95% confidence intervals:** we resample the evaluation items 2,000 times, using the same resampled items for every seed and condition (a paired design).
- **Why over items, not seeds:** 3 seeds is too few for seed-level tests, and item variability is the main source of noise.
- **Sign consistency:** do all 3 seeds point the same way?
- **McNemar test:** compares right/wrong flips on the same items.
- **Honest caveats:** about 20 comparisons per model, no multiple-testing correction, and only 3 seeds. So this is a **feasibility sketch, not proof**.

3.15 The McNemar test, explained

	Gray image: right	Gray image: wrong
Real image: right	83 (right both times)	**144** (the image helped)
Real image: wrong	**49** (the image hurt)	224 (wrong both times)

- It compares **two conditions on the same items** (Qwen, 500 MMStar items, real vs gray image).

- Items that came out the same both times (83 and 224) tell us nothing, so it ignores them and looks only at the 193 that changed.
- If the image didn't matter, the changes would split like coin flips (about 96 each way). We got **144 vs 49**: p = 4.9e-12.
- **Why not a normal two-proportion test:** that assumes independent groups. These are the same questions, so a paired test is the right one.

4. Results and how to explain them

4.1 Tier 1: the measurement works

Model	Real	Gray	Swap
Qwen2-VL	45.4%	26.4%	23.0%
SmolVLM	40.0%	20.4%	23.4%
InternVL3	56.8%	30.2%	31.2%

Chance is 25%. For Qwen, McNemar gives 144 items flipping right-to-wrong without the image and 49 the other way, p = 4.9e-12.

4.2 Main result: text90 minus text10 (points, 95% CI)

Measure	Qwen2-VL	SmolVLM	InternVL3
Overall accuracy (A)	**-2.6** [-4.3, -1.0] *	-0.3 [-2.1, +1.5]	-0.5 [-2.5, +1.3]
G accuracy	-1.3 [-3.0, +0.4]	-1.1 [-2.9, +0.5]	**-2.6** [-4.5, -0.7] *
G benefit, real - gray	**-2.5** [-4.6, -0.2] *	-1.6 [-3.7, +0.5]	**-3.7** [-6.1, -1.3] *
G benefit, real - swap	-1.6 [-4.0, +0.6]	-1.8 [-4.1, +0.3]	**-4.2** [-6.9, -1.7] *
Seeds agreeing (real - gray)	3/3	3/3	3/3

* = the confidence interval excludes 0. A negative number means the 90% text-answerable run did worse.

- **InternVL3** is the clean example of a hidden regression: accuracy flat, image use down, 3/3 seeds.
- **Qwen:** image benefit drops, but overall accuracy also drops. **Useful detail:** splitting the A set shows Qwen's drop sits on the **image-needing** items (text90: 42.1% to 39.1%; text10: 42.1% to 42.6%), while the text-guessable items barely changed (58.1% vs 58.8%). So it's still a visual-grounding regression; it just shows up in overall accuracy too.
- **SmolVLM:** same direction, but no confidence interval excludes zero.

4.3 Paired-image tests: same answer to both images (text90 minus text10)

Model	MMVP	NaturalBench
Qwen2-VL	**+4.2** [+1.1, +7.6] *	+1.5 [-0.5, +3.5]
InternVL3	+2.0 [-3.1, +6.9]	+1.3 [-0.9, +3.4]
SmolVLM, original mixes	**-4.7** [-8.9, -0.4] * (reversed)	**-4.6** [-7.0, -2.2] * (reversed)
SmolVLM, yes/no-balanced mixes	**+7.8** [+2.2, +13.3] *	**+15.4** [+12.5, +18.5] *

Positive = the 90% run gave the same answer to both images more often, i.e. it ignored the image more (the predicted direction).

4.4 Why the effects are small (2-4 points)

- Training was short (about 1 epoch), small (1,000 items) and gentle (LoRA).
- That fits *Fine-tuning MLLMs Without Forgetting*, which found LoRA keeps changes within about ±3 points.
- Small but consistent is what we needed. A forecaster needs a target that moves measurably, not a big one.

4.5 The yes/no story (SmolVLM's paired tests)

- On the original mixes, SmolVLM's paired tests went the **opposite** way to the prediction.
- Investigation showed the mixes differed in yes/no balance (75% vs 36% "yes"). The text10 runs said "Yes" 80% of the time, versus 50% truth.
- We rebuilt the mixes with VSR at **50% yes** (900 items) and retrained SmolVLM, costing 3.6 GPU hours.
- The result flipped to the predicted direction: **+7.8 on MMVP, +15.4 on NaturalBench**, 3/3 seeds.
- **But** those runs said "Yes" 90% of the time and picked option A 75% of the time. So for SmolVLM, the paired tests mostly track **answer bias**.
- **Lesson:** training data must balance **answer types**, not only sources, and same-answer metrics need a check for answer bias. **Say this proudly:** we found a confound, tested it, and reported it honestly.

5. Limitations (say these before they ask)

- Only 3 seeds, about 20 comparisons per model, and no multiple-testing correction.
- Only small (~2B) models.
- Labels and the G set come from Qwen and were reused for the other models.
- One training setting: one learning rate, one rank, about 1 epoch.
- Mixes were matched on source but at first not on answer type. Only SmolVLM was retrained on balanced mixes.
- **No forecaster yet, and only 18 runs.** That is far too few to train one.

6. The forecaster (next phase): what to say

- **Target:** the final change in image benefit on the G set compared with step 0 (ΔG).
- **Inputs:** static dataset features measured before training (loss-gap level, source balance, answer balance) and early training signals (per-class loss, gradient norm, image attention, early checkpoint scores).
- **Model:** a gradient-boosted tree (XGBoost or LightGBM) or a small MLP. SHAP can explain which feature drove a prediction.
- **Data needed:** many more runs, varying mix ratios, learning rate, rank and seeds. Each run is one training example for the forecaster.
- **Baselines to beat:** static features only; extrapolating the early loss curve; the step-50 checkpoint score.
- **Closest prior work: TuneAhead** (ICML 2026) uses the same static + early-probe design to predict **accuracy** in text models. Our difference: we predict **image reliance** in **VLMs**, which accuracy-based predictors miss. Say openly that the design is adapted from TuneAhead.
- **Output:** a risk level, so that a high-risk run can be stopped early or its data rebalanced.

7. Likely questions and answers

"Why is this a problem if accuracy doesn't drop?"

Because accuracy is measured on benchmarks where shortcuts often work. A model that stopped looking fails on unusual real cases. *Seeing without Looking* shows benchmarks don't catch this.

"Isn't 2-4 points too small to matter?"

It's small but consistent: the CIs exclude zero and all seeds agree. It came from only about 1 epoch of gentle LoRA training. For a pilot, what matters is that it's measurable. Longer or more aggressive training would likely give bigger effects, but that's untested.

"Why gray images? Isn't that unnatural?"

Yes, which is why we also use swapped images (a real but wrong picture). Both give the same picture of the results, and InternVL3 drops on both.

"Why didn't you fine-tune the vision encoder?"

To keep the image features fixed. Then any change must come from how the language model uses them. It's also the standard LoRA practice.

"Why only 3 seeds?"

Compute: 18 runs on one laptop GPU. To make up for it, we run statistics over items (bootstrap, McNemar) and check that seeds agree in direction.

"Did you correct for multiple comparisons?"

No, and we say so. These results are exploratory and labelled a feasibility sketch.

"Why did Qwen's overall accuracy also drop?"

The drop sits on the image-needing part of the A set (42.1 to 39.1), not the guessable part. So it's the same visual regression, just visible in overall accuracy for this model.

"Why does SmolVLM behave differently?"

Different architecture and a strong default-answer bias that's very sensitive to the yes/no balance of the mix. Its MMStar results point the same way but aren't significant.

"Could the effect come from source datasets rather than text-answerability?"
No. Both mixes have exactly 250 items from each source, and the thresholds are set within each source.

"Could training data overlap with the test set?"
No. The hash and text guard checks against all 1,500 MMStar items plus MMVP and NaturalBench, stops on any overlap, and is tested with a planted duplicate.

"Why letter probabilities instead of generated answers?"
They're deterministic, need no answer parsing, take one forward pass and give a confidence. It's a standard way to evaluate multiple choice.

"Why MMStar?"
It was designed so questions need the image, and our blank test confirms it: 26% is about chance.

"How is this different from TuneAhead?"
TuneAhead predicts accuracy for text LLMs. We target the hidden loss of visual grounding in VLMs, which accuracy doesn't show. We adapt their static + dynamic feature idea to a multimodal target.

"How is this different from 'Fine-tuning MLLMs Without Forgetting'?"
They show LoRA keeps accuracy stable, and we agree: our A is flat too. But we show image reliance can still drop underneath, which they didn't measure.

"Why use Qwen's labels for all models?"
So all three models train on the identical data and only the model differs. The trade-off is that the labels reflect Qwen's view. Model-specific labels are an extension.

"How many runs will the forecaster need?"
Many more than 18; at least dozens. TuneAhead used over 1,300 runs, but on bigger hardware. We'd start small, using shorter runs and varied mixes, and report learning curves.

"Why should we believe the forecaster will work?"
We don't claim it will yet. The pilot shows the target is measurable and moves with the data mix, which is the precondition. TuneAhead shows that early signals predict outcomes for accuracy.

"What is LoRA, and what does rank 16 mean?"
LoRA freezes the original weights and learns a small correction next to each layer, as the product of two thin matrices. The rank is the width of that bottleneck: with rank 16, each correction passes through 16 dimensions. It's a standard setting, about 18 million trainable numbers, kept the same in every run so it can't be a confound. We did not test other ranks.

"What hardware did you use?"
One RTX 4060 Laptop GPU with 8 GB. Everything ran locally with a heat guard, disk guard and resumable jobs.

"What was the hardest problem?"
The yes/no confound. Our paired-test result reversed, we traced it to answer balance, rebuilt the mixes, retrained, and found the metric itself was confounded by answer bias.

8. Related work, ranked

Rank	Paper	Why it matters to us
1	TuneAhead (Luo et al., ICML 2026)	Closest to our main goal: forecasts fine-tuning results from static features + early probe signals. Predicts accuracy for text LLMs; we target image reliance in VLMs.
2	Fine-tuning MLLMs Without Forgetting (Li, Zhang et al., 2026)	Same setting (LoRA on VLMs) but measures only accuracy, which is the gap we fill.
3	Seeing without Looking (Lan et al., 2026)	Accuracy hides weak use of the image; we extend this to fine-tuning.
4	Words or Vision (Deng et al., CVPR 2025)	VLMs trust text over images, linked to the training data mix; includes Qwen2-VL.
5	VisNec (Dong et al., 2026)	Visual-necessity score is essentially our loss gap; they select data, we vary the mix.
6	IIMM (Niss et al.)	Forecasting precedent, but for CLIP-style models and accuracy.

Rank	Paper	Why it matters to us
7	Dancette et al. (ICCV 2021)	Classic background: shortcuts hide behind accuracy.
8	Learning Dynamics of VLM Finetuning (Zhang et al., 2025)	Weak fit (DPO alignment); one sentence on probe signals for early stopping.

Also cite the datasets and models you used: MMStar, MMVP, NaturalBench, The Cauldron, LoRA, QLoRA, Qwen2-VL, SmolVLM, InternVL3.

9. Numbers cheat sheet

Item	Value
Training pool	3,161 items (4 subsets); 33 duplicates and 6 overlaps removed
Mixes	1,000 items each (250 per subset); 10% vs 90% text-answerable; mean gap 1.17 vs 0.09
Balanced follow-up	900 items, VSR at 50% yes, 3.6 GPU hours (SmolVLM only)
Evaluation sets	A = 500; G = 581; MMVP = 150 pairs / 300 questions; NaturalBench = 298 samples / 1,192 questions
LoRA	rank 16, alpha 32, LR 2e-4, batch 8, 120-125 steps, ~18M trainable parameters (<1%)
Runs	18 (3 models x 2 mixes x 3 seeds), 12-21 min each, 2.1-5.5 GB peak memory
Hardware	RTX 4060 Laptop GPU, 8 GB
Bootstrap	2,000 resamples over items
Tier 1 (Qwen)	45.4% real vs 26.4% gray; McNemar 144 vs 49, p = 4.9e-12
Headline	InternVL3: A -0.5 (n.s.); G accuracy -2.6; G benefit -3.7 / -4.2; 3/3 seeds